Case Study: Simulated Drone Delivery System

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1 Team Information

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2 Introduction

A simulated drone delivery system is an Artificial Intelligence (AI) application in which autonomous drones transport packages from source locations to specified destinations. The system operates within a simulated environment containing delivery locations, obstacles, restricted areas, charging stations, and changing environmental conditions.

The proposed system is designed as a **multi-agent environment** in which multiple autonomous Drone Agents operate simultaneously. A Delivery Manager Agent assigns delivery requests to suitable drones, while a Weather Agent introduces changing environmental conditions that can affect navigation. Each Drone Agent perceives its surroundings, plans a route, monitors its battery, avoids obstacles, and delivers its assigned package.

The delivery problem is modeled as a state-space search problem. Each drone selects an efficient route while considering distance, battery usage, obstacles, and environmental conditions. **A* Search** is used for route planning, and dynamic replanning allows drones to respond when the environment changes.

3 PEAS Formulation

PEAS stands for **Performance Measure, Environment, Actuators, and Sensors**. It provides a structured description of the task environment in which the intelligent agents operate, including their objectives, surroundings, available actions, and the information they receive through their sensors. For the simulated drone delivery system, PEAS defines how the Drone Agents perceive and interact with the delivery environment.

Component	Description
Performance Measure	Successful and timely package delivery, minimum travel distance, efficient battery usage, collision avoidance, safe navigation, and effective use of available drones.
Environment	A simulated grid-based delivery area containing drone starting locations, delivery locations, packages, obstacles, restricted cells, charging stations, and changing weather conditions.
Actuators	Movement actions such as moving up, down, left, right, hovering, picking up packages, delivering packages, and returning to a charging station.
Sensors	Position, destination, obstacle detection, battery level, package status, and current weather information.

4 Environment & Agent Analysis

4.1 Environment Characteristics

The simulated drone delivery environment has the following characteristics:

- **Partially Observable:** A drone may not have complete information about future environmental changes and relies on available map and sensor information.
- **Dynamic:** Weather conditions, drone positions, and available routes can change during execution.
- **Sequential:** Each movement affects the drone's subsequent position, battery level, and remaining route.
- **Stochastic:** Environmental conditions such as weather can change during the simulation.
- **Discrete:** The delivery area is represented as a grid in which each cell represents a possible drone location.
- **Multi-Agent:** Multiple autonomous Drone Agents operate in the same environment and receive different delivery tasks.
- **Known:** The basic map and movement rules are known, while dynamic conditions such as weather can change.

4.2 Agents in the System

The system contains three types of autonomous software agents.

1. Drone Agent

The Drone Agent is the primary agent responsible for executing deliveries. It navigates to the assigned package, picks it up, plans a route to the destination, avoids obstacles, monitors its battery, and delivers the package.

If the battery becomes low, the drone can travel to a charging station before continuing its delivery. If the environment changes, the drone can calculate a new route.

2. Delivery Manager Agent

The Delivery Manager Agent coordinates delivery requests and assigns packages to available drones. The assignment considers factors such as distance to the package, battery level, workload, and drone availability.

The manager performs task allocation but does not directly control the movement of the drones.

3. Weather Agent

The Weather Agent models changing environmental conditions.

Weather	Effect
Clear	Normal movement cost
Windy	Increased movement/energy cost
Rain	Higher movement cost and restricted cells
Storm	Selected routes may become unavailable

The Weather Agent changes environmental conditions that are perceived by the Drone Agents.

Battery is not a separate agent. It is an internal state variable of the Drone Agent. Packages, obstacles, and charging stations are environmental entities.

5 Algorithmic Modeling & Search Strategy

5.1 State-Space Formulation

The drone delivery problem is formulated as a state-space search problem:

- **Initial State:** A drone is positioned at its starting location with an assigned delivery task.
- **Actions:** Move to an adjacent grid cell, pick up a package, deliver a package, or move towards a charging station.
- **Transition Model:** An action changes the drone's position, battery level, package status, and route.
- **Goal Test:** The package is successfully delivered to its assigned destination.
- **Path Cost:** The cost is primarily based on travel distance, with additional penalties for environmental conditions when required.

The environment is represented as a graph where valid grid cells are nodes and movement between adjacent cells represents an edge.

5.2 A* Search for Route Planning

Each Drone Agent uses **A* Search** to find an efficient route from its current position to the destination.

The evaluation function is:

$$f(n) = g(n) + h(n)$$

where $g(n)$ is the cost from the starting node to node n, and $h(n)$ is the estimated cost from node n to the destination.

For the grid-based environment, Manhattan distance is used as the heuristic:

$$h(n) = |x_n - x_g| + |y_n - y_g|$$

where (x_n, y_n) represents the current position and (x_g, y_g) represents the destination.

A* is selected because it considers both the cost already incurred and the estimated remaining cost. It can therefore find an efficient path while avoiding obstacles and restricted cells.

5.3 Task Allocation

The Delivery Manager Agent assigns delivery requests to available drones.

For each delivery request, the manager considers:

- Distance to the package.
- Current battery level.
- Current workload.
- Drone availability.

The drone with the most suitable overall conditions is assigned the delivery. This allows multiple drones to share the delivery workload.

5.4 Dynamic Replanning

The environment can change while a drone is travelling. For example, rain or a storm may make a previously selected route unavailable or more costly.

When a significant environmental change is detected, the Drone Agent:

1. Detects the change.
2. Checks whether the current route is still suitable.
3. Runs A* again if necessary.
4. Continues the delivery using the new route.

This allows the system to adapt to dynamic conditions rather than following a fixed route.

5.5 Search Strategy Selection

Several search algorithms can be considered for the drone routing problem. **Breadth-First Search (BFS)** can find a shortest path when all movements have equal cost, but it does not consider the estimated distance to the destination. **Depth-First Search (DFS)** may find a path quickly but does not guarantee an optimal solution. **Uniform Cost Search (UCS)** handles different path costs, but it does not use heuristic information to guide the search toward the destination.

A* Search is selected as the primary search strategy because it combines the actual path cost with a heuristic estimate of the remaining distance. This allows the drone to efficiently explore promising routes while avoiding obstacles and restricted cells. The Manhattan distance heuristic is particularly suitable for the proposed grid-based environment.

A* can also be executed again when the environment changes. Therefore, if weather conditions make the current route unavailable or more costly, the Drone Agent can perform route replanning from its current position.

The **Delivery Manager Agent** performs task allocation, while the **Weather Agent** introduces environmental changes. The **Drone Agents** use A* for navigation and replanning, allowing the multi-agent system to coordinate delivery tasks while adapting to changes in the environment.